

Contents

ACRME Architecture Decision Records (ADRs)	1
About This Document	1
ADR-001 — Region Selection	2
Context	2
Decision	2
Consequences	3
Alternatives Considered	3
ADR-002 — Quota and Capacity Management	4
Context	4
Decision	4
Consequences	5
Alternatives Considered	5
ADR-003 — Capacity Management during Disaster Recovery (DR)	6
Context	6
Decision	6
Consequences	7
Alternatives Considered	7
ADR-004 — Forecast and Increase of Capacity and Quota	8
Context	8
Decision	8
Consequences	9
Alternatives Considered	9
Appendix A — Decision Log Cross-Reference	9
Appendix B — Status Legend	10
Appendix C — Evidence Tag Taxonomy	10

ACRME Architecture Decision Records (ADRs)

Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect — Architecture Governance
Version: 1.0
Date: August 2026
Status: Accepted

About This Document

An **Architecture Decision Record (ADR)** captures a single significant architectural decision, the context that forced the decision, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original.

This document records the four foundational ACRME architecture decisions:

ADR	Title	Status	Supersedes
ADR-001	Region Selection	Accepted	—
ADR-002	Quota and Capacity Management	Accepted	—

ADR	Title	Status	Supersedes
ADR-003	Capacity Management during Disaster Recovery (DR)	Accepted	—
ADR-004	Forecast and Increase of Capacity and Quota	Accepted	—

Evidence tags used throughout: **[Documented]** (Azure platform docs), **[Decided]** (Decision Log D1–D11), **[Derived]** (logical consequence), **[Assumed]** (architectural judgement pending validation).

ADR-001 — Region Selection

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, Platform Engineering, DR Owner

Related decisions: D1, D4, D5, D8, D9; HC-1..HC-10; VR-1..VR-11

Source: multi_region_placement_design.md §27–28; acrme_production_readiness_review_and_architecture §27

Context

Customers request capacity by supplying either a specific Azure region or an Azure geography (e.g. “US”, “Europe”, “Middle East”). The engine must derive three environment regions — **Prod**, **NonProd/CVAL**, and **DR** — that satisfy isolation, resilience, capacity, quota, and data-residency requirements.

Forces at play:

- Azure regions differ in SKU availability, zone count, capacity headroom, and quota posture. A naive “pick the nearest region” approach silently produces single-zone or capacity-starved placements.
- Some regions are **Restricted Capacity Regions** (sovereign clouds, limited-SKU regions) that must never be selected by automated placement.
- Certain geographies (notably the **Middle East**, with only UAE North and Saudi Arabia Central) do not have enough Standard regions to satisfy a three-region model in-geo, forcing a governed cross-geo DR path.
- Placement decisions must be **deterministic, auditable, and replayable** — the same inputs must always produce the same output, and every decision must be reconstructable for compliance.

Decision

Adopt a **staged, constraint-then-score placement pipeline** driven by environment-type-specific scoring formulas:

1. **Stage 1 — Eligibility pre-filtering.** Reduce all Azure regions to Standard Capacity Regions within the customer’s geography.
 - **HC-9 STANDARD_REGION_ONLY** excludes Restricted regions before scoring. **[Decided – D9]**
 - **HC-8 GEOGRAPHY_CONTAINMENT** confines the Prod anchor to the customer’s chosen geography. **[Derived]**

2. **Stage 2 — Hard-constraint gate.** Each surviving region must pass **HC-1..HC-7 and HC-10** (region separation, capacity floor, quota floor, DR separation class, zone availability, DR coverage floor, DR floor integrity, cross-geo extension approval). Any failure excludes the region from scoring entirely — hard constraints are binary gates, never scored penalties. [Decided - D4]
3. **Stage 3 — Environment-type-specific scoring.** Rank survivors using three formulas sharing default weights ($\alpha=0.30$, $\beta=0.20$, $\gamma=0.25$, $\delta=0.15$, $\epsilon=0.10$) but with env-type-specific semantics for (capacity signal) and (DR readiness): [Decided - D9]
 - **Prod:** $\text{Prod_region} = \text{argmax}(\text{PS_Prod}(r))$ over eligible Standard regions in-geo.
 - **NonProd/CVAL:** $\text{argmax}(\text{PS_NonProd}(r))$ over survivors, excluding the Prod region.
 - **DR:** $\text{argmax}(\text{PS_DR}(r))$ over survivors, excluding the Prod region (but **may** share with Non-Prod per D8).
4. **Selection order is sequential — Prod → NonProd → DR** — not joint optimization. With 3–4 regions the greedy sequential result equals the joint-optimal result in nearly all practical cases, while remaining transparent and auditable. [Decided - D1]
5. **Middle East special handling.** Prod and NonProd are chosen in-geo via $\text{argmax}(\text{PS_Prod})$ over {UAE North, Saudi Arabia Central}; DR is assigned to **Belgium Central** via the only approved Cross-Geo Extension paths (Saudi Arabia → Belgium Central, UAE North → Belgium Central). Belgium Central must itself pass HC-1..HC-10; if it fails, the placement is rejected with an ops alert — the engine never silently substitutes another region. [Decided - D8; Documented - §27]
6. **Determinism & audit.** All candidate sets, sub-scores, the winning score, and the active PlacementPolicy version are written to the OperationRecord for replay (VR-8, VR-9). Even the 3-region edge case (single eligible candidate) runs the full scoring path so the score is logged as a capacity-health signal. [Decided - D5]
7. **Operating mode.** In Phase 1 the engine runs region selection in **recommendation/shadow mode** — it produces and logs the ranked recommendation but does not autonomously place. Autonomous placement is gated on POC validation of the scoring formulas.

Consequences

Positive: - Deterministic, replayable placement with a complete audit trail satisfies compliance and post-incident review. - Hard constraints guarantee no placement ever violates isolation, zone, quota, or residency rules regardless of score. - Restricted regions are structurally impossible to select automatically; exceptions flow through a governed Scenario 2 workflow with a persisted, revocable approval ID. - Env-type-specific formulas let Prod optimise for capacity/isolation while DR optimises for overflow readiness.

Negative / trade-offs: - Three formulas plus 16 per-CRG-type RegionalSnapshot fields increase implementation and snapshot-maintenance cost (backlog E07-S16, E03-S10). - Sequential selection is greedy; if the region count ever exceeds 6, joint optimization should be revisited (D1 review trigger). - Cross-geo extension paths are a manual governance artefact — adding a path requires a PlacementPolicy update, governance approval, and a Decision Log entry. - Worked scoring examples remain a known gap (G-7) until per-CRG-type inputs are finalised.

Neutral: - CRG_Score (RCW) is demoted from a primary scoring input to a monitoring-only signal (D9).

Alternatives Considered

Alternative	Why Rejected
Joint optimization (select all three regions simultaneously)	Higher combinatorial complexity; opaque and harder to audit; identical result to sequential for 3–4 regions. [D1]
Single generic PS(r, env_type) formula	and have fundamentally different meanings per env type; a single formula needs so many branches it becomes three formulas anyway. [D9]
Geographic distance as a scored soft objective	Operators already choose geographically independent region sets at design time; HC-4 as a hard constraint is sufficient and simpler. [D4]
Silent fallback region for failed Middle East DR	Violates governance and residency guarantees; the engine must fail loudly and require an approved extension path. [S27]

ADR-002 — Quota and Capacity Management

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, Platform Engineering, FinOps

Related decisions: D6, D7, D9; HC-2, HC-3, HC-7

Source: multi_region_placement_design.md §Quota Groups; design_change_summary.md QG-1..QG-12

Context

Each customer needs three CRGs per region (Prod, NonProd, DR). Azure tracks compute quota per subscription per region per SKU family; quota exhaustion silently blocks CR creation and VM deployment. The engine must:

- Prevent a NonProd surge from consuming quota that Prod CR creation needs (isolation).
- Make emergency capacity transfer (ADR-003) **quota-neutral** where possible — i.e. reshuffle capacity without waiting hours for an Azure quota-increase approval during a crisis.
- Provide clean per-environment cost attribution for chargeback.
- Enforce a protected DR reservation that NonProd growth can never erode.

Azure **Quota Groups** (Preview) allow multiple subscriptions/CRGs to share a group-level quota pool, but Azure does **not** natively support intra-group sub-reservations — so any sub-partition of a group pool must be enforced by the engine.

Decision

Adopt a **Two-Quota-Group-per-region model with an engine-enforced DR floor**:

1. **Two groups per region:** one **Prod-only** group and one **shared NonProd+DR** group. [Decided - D6]

Prod_Group_Limit(R) = base_subscription_quota_limit × (1 + emergency_transfer_headroom)

NonProd_DR_Group_Limit(R) = base_subscription_quota_limit × (1 + emergency_transfer_headroom)

2. **Engine-enforced DR floor** inside the NonProd+DR group (Azure has no native sub-reservation): [Decided - D7]

`DR_Floor_vCPU(R)` = `potential_dr_demand(R) × vCPU_per_instance × dr_ratio_max`
`Effective_NonProd_Ceiling(R)` = `NonProd_DR_Group_Limit(R) - DR_Floor_vCPU(R)`

The floor is always sized at **dr_ratio_max (0.40)** — the upper bound of the DR ratio range — so it never needs to grow if the operating ratio is later raised (0.30 → 0.40). Only actual Prod demand growth expands the floor. [Decided - D7]

3. Hard-constraint enforcement at placement:

- **HC-2 CAPACITY_FLOOR** — CR headroom $2 \times \text{requested_vm_count} \times \text{SKU_vCPU}$.
- **HC-3 QUOTA_FLOOR** — env-type-aware group headroom check (Prod group / effective NonProd ceiling / DR headroom). Reads from the `QuotaGroup` entity, not legacy per-subscription `QuotaRecord`. [Decided - D9]
- **HC-7 DR_FLOOR_INTEGRITY** — rejects NonProd placement that would push `nonprod_quota_used` above the effective ceiling, evaluated **after** HC-3. [Decided - D7]

4. **Legacy per-subscription quota tracking is retained for monitoring/audit only** and superseded for all placement and scoring checks by group-level fields. [Decided - D6]

5. **Continuous protection:** the reconciliation loop watches for floor violations and emits `DRFloorViolationDetected`; NonProd CRG scale operations that approach the ceiling are operator-gated.

Worked POC topology (GP-06): Prod group 128 vCPU; NonProd+DR group 80 vCPU; DR floor 32 vCPU → effective NonProd ceiling 48 vCPU.

Consequences

Positive: - Prod isolation is enforced at the Azure control-plane level — a NonProd surge cannot starve Prod. - Tier 2 Emergency Capacity Transfer becomes **truly quota-neutral** (ARM operations only, no Azure approval gate) because NonProd and DR draw from the same group pool (see ADR-003). - Clean chargeback: one Azure billing record per group per region. - The DR floor guarantees a protected reservation NonProd can never erode.

Negative / trade-offs: - The DR floor is a **soft ceiling the engine must maintain continuously** — Azure won't enforce it. A bug could allow NonProd to breach the floor; mitigated by `DRFloorViolationDetected` alerting and operator gates. [D6] - Sizing the floor at `dr_ratio_max` over-reserves quota slightly at lower operating ratios — a deliberate reliability-over-cost trade, reviewed quarterly. [D7] - `Quota_Score` and HC-3 must be environment-type-aware (three code paths). - **Hard dependency on Azure Quota Groups GA** — Blocker B-1 (POC-30); if `groupQuotas` returns 404 in the target tenant/region, escalate to Azure Support.

Alternatives Considered

Alternative	Why Rejected
Single shared quota pool (all three CRGs)	Breaks Prod isolation — a NonProd surge can consume Prod's quota. [D6]
Three separate groups (one per CRG)	Maximum isolation but makes Tier 3 transfer non-atomic — releasing NonProd quota lands in the wrong group, forcing an Azure quota request mid-crisis (hours of RTO impact). [D6]

Alternative	Why Rejected
Per-subscription quota only (legacy)	Cannot make emergency transfer quota-neutral without risky cross-subscription coordination. [D6]
DR floor sized at current ratio	Undersizes the floor if the ratio is later increased, blocking DR expansion. [D7]

ADR-003 — Capacity Management during Disaster Recovery (DR)

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, DR Owner, Platform Engineering, Security

Related decisions: D8, D10, D11; HC-1, HC-4, HC-6; G-14, G-15

Source: multi_region_placement_design.md §Emergency Capacity Transfer; acrme_production_readiness_review §29

Context

When a primary region fails, customers must recover in their DR region within a committed RTO. Pre-positioning a full 1:1 DR reserve is prohibitively expensive, yet under-provisioning risks a failed failover. During a crisis, waiting on Azure quota-increase approvals (hours) is unacceptable. The engine must also strictly separate **routine growth** from **destructive crisis operations** so that reconciliation can never accidentally trigger VM disruption.

Two capabilities are needed: - A way to reuse otherwise-idle NonProd capacity as DR overflow. - A safe, tiered escalation model for expanding DR capacity during a declared event.

Decision

Adopt **NonProd/DR co-location with a coverage floor**, a **formal two-mode engine state**, and a **three-tier emergency transfer model**:

1. **NonProd and DR may share a region (HC-1 constraint removed)**. This lets the NonProd CRG's `effective_free` (after `dr_overflow_reserve`) count as DR overflow headroom. [Decided - D8]

- **HC-6 DR_COVERAGE_FLOOR** guarantees the combined pool can absorb demand before DR placement is accepted:

$$\text{dr_crg_free_slots}(R) + \text{nonprod_crg_effective_free}(R) \geq \text{prod_vm_count} \times \text{dr_ratio_max}$$

- **HC-4 DR_SEPARATION_CLASS** still guarantees Prod and DR sit in non-correlated failure domains (HIGH separation for non-paired regions).

2. **Two separate operating systems gated by engine_mode**: [Decided - D10]

- **STEADY_STATE** — organic growth via the reconciliation loop and `CapacityIncreaseRequest` (ADR-004). Emergency Transfer is **rejected** in this mode.
- **DR_EVENT_ACTIVE** — crisis operations only. The auto-increase trigger is **suppressed** in this mode. Mode transitions are operator-gated with dual approval (state machine `EngineModelState`, PRR §29), never automatic. [Decided - D10]

3. Three-tier Emergency Capacity Transfer escalation: [Decided - D11]

- **Tier 1 — DirectExpansion (automated):** expand DR CR quantity using free headroom in the DR group. No approval beyond DR-event declaration.
- **Tier 2 — QuotaNeutralTransfer (policy-gated):** reduce a NonProd CR (releasing quota to the shared NonProd+DR group pool) and expand the DR CR from that same pool. Net group headroom change 0 — **quota-neutral, no VM execution-state change** (only Non-Prod SLA is removed). This is possible *only* because of the two-group model (ADR-002).
- **Tier 3 — DestructiveTransfer (dual approval + elevated RBAC):** the only tier that modifies VM-to-CRG associations. The operator supplies an explicit `vm_disassociation_list` (no automated VM selection in Phase 1); executed via 6-step Path B with Path A fallback per VM. VMSS entries are rejected in Phase 1.

4. Quota-neutral math (Tier 2):

NonProd CR reduction → releases quota to shared NonProd+DR GROUP pool
DR CR expansion → consumes from the SAME pool
net group headroom change 0 (ARM operations only; no Azure quota approval)

5. **DR reserve sizing** is 30-40% of Prod (`dr_ratio_min=0.30`, `dr_ratio_max=0.40`, `dr_ratio_target=0.35`); the protected floor uses `dr_ratio_max` (ADR-002/D7).
6. **Phase-1 safety posture:** Tier 2 is approval-gated; **Tier 3 is blocked** pending the G-14 consumer-credential model and G-15 engine-mode state machine. No invented SLA — propagation/approval times are measured and reported as unknown until observed.

Consequences

Positive: - Idle NonProd capacity doubles as DR overflow, cutting the cost of pre-positioned DR reserve while HC-6 guarantees sufficiency. - The `engine_mode` gate makes it structurally impossible for routine reconciliation to trigger destructive VM operations. - Tier 2 delivers meaningful crisis capacity **with zero VM disruption and zero Azure quota wait** — the common escalation path avoids Tier 3 entirely. - Every operation is tagged with its operating mode for a clean audit trail.

Negative / trade-offs: - Co-location adds risk that NonProd over-consumption erodes DR overflow; mitigated by HC-6, `dr_overflow_reserve`, and floor alerting. - `engine_mode` must be a formal Cosmos DB state propagated to every reconciliation cycle, with operator-gated, dual-approval transitions. - **Tier 3 is blocked** until G-14 (Managed Identity vs cross-tenant SP credential model) and G-15 (state machine) are resolved — a known Phase-1 limitation. - Tier 3 requires a rare, audited break-glass role (`ACRME.SuperAdmin`) for single-approver override. - Requires per-CRG-type `RegionalSnapshot` fields to evaluate HC-6.

Alternatives Considered

Alternative	Why Rejected
Keep DR_region NonProd_region	Prevents using NonProd as DR overflow — the whole point of co-location. [D8]
Remove separation with no compensating HC	Risks NonProd over-consumption leaving no DR headroom. [D8]
Unified capacity engine with mode flags	Mode flags create complex conditionals and risk triggering VM disassociation during routine reconciliation. [D10]

Alternative	Why Rejected
Emergency transfer as an extension of auto-increase	Conflates policy-driven growth with operator-gated crisis response; could run destructive ops without a DR declaration. [D10]
Two tiers only (automated + destructive)	Loses the quota-neutral Tier 2 — the critical low-risk intermediate that avoids Tier 3 in most crises. [D11]
Four tiers (separate VMSS tier)	VMSS disassociation deferred as a Phase-1 limitation rather than a distinct tier. [D11]

ADR-004 — Forecast and Increase of Capacity and Quota

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, Platform Engineering, FinOps, Capacity Planning

Related decisions: D10; FR-7, FR-4.4; G-24

Source: `azure_cr_management_engine_design.md` FR-7; `acrme_production_readiness_review_and_architecture` §Forecast/§30

Context

Reserved capacity must grow ahead of demand — but Azure quota increases take time to approve, and over-reservation wastes money. The engine needs to forecast demand, recommend right-sized capacity, and drive quota/CR increases with enough lead time, **without** ever autonomously performing material or destructive changes in Phase 1. This growth path must be architecturally distinct from crisis operations (ADR-003).

Decision

Adopt **forecast-driven, approval-gated capacity growth** operating exclusively in `STEADY_STATE`:

1. **Forecasting** analyses historical CR allocation and projects demand over a configurable window (default 30/60/90 days), exposing raw time series and derived recommendations via API. [Documented - FR-7.1/7.6]

2. **Capacity sizing formula:** [Documented - FR-7.3]

$$\text{Forecast_Quantity} = \text{ceil}(\text{Forecast_Peak} \times (1 + \text{Growth_Buffer}) + \text{DR_Buffer})$$

Increase when forecast demand exceeds current reserved quantity within the window; right-size down when demand is consistently below current, honouring a configurable buffer.

3. **Lead-time alerting:** when forecast demand approaches a quota limit (default **80%**), emit `ForecastApproachingQuotaLimit` with a **14-day lead time** — enough for quota-increase processing. [Documented - FR-7.4]
4. **Auto-increase is approval-gated in Phase 1.** The trigger uses utilisation thresholds with **debounce/cooldown** to avoid thrashing; a `CapacityIncreaseRequest` entity carries the full lifecycle (create → approve → execute → retry → cancel), and Phase 1 requires **operator approval** before execution. [Decided - D10; Derived - §30, G-24]

5. **Quota increases** are initiated via the Azure Support REST API (`Microsoft.Capacity/.../serviceLimits`) — at group level where Quota Groups apply (ADR-002) — subject to the same operator-approval gate. [Documented - FR-4.4]
6. **Mode isolation.** Auto-increase runs **only** in `STEADY_STATE` and is **suppressed during `DR_EVENT_ACTIVE`**, keeping organic growth strictly separate from crisis transfer (ADR-003). [Decided - D10]
7. **Non-destructive guarantee.** Increases only ever raise CR quantity or request more quota; guarded reduction (right-sizing) never drops a CR below its allocated count (platform floor, FR-1.6) and is itself approval-gated.

Consequences

Positive: - Capacity grows ahead of demand with sufficient lead time for quota approvals — reducing capacity-exhaustion incidents. - The debounce/cooldown and approval gate prevent runaway or thrashing increases. - `CapacityIncreaseRequest` gives a fully auditable, retryable, cancellable growth workflow. - Right-sizing recovers cost from over-reserved CRs without risking allocated VMs.

Negative / trade-offs: - Approval-gated in Phase 1 means growth is not instantaneous — acceptable because Emergency Transfer (ADR-003) covers crisis speed. - Forecast accuracy depends on history; workload-tagged per-workload forecasts (FR-7.5) are only partially covered (gap G-16). - Threshold, buffer, and cooldown values are empirical and require tuning during the pilot; SLA/propagation times are measured, not invented. - `CapacityIncreaseRequest` lifecycle (entity, approval, retry, cancellation) is still a backlog item (G-24) requiring an end-to-end approved-increase test.

Alternatives Considered

Alternative	Why Rejected
Fully autonomous auto-increase	Removes human control over material cost/quota changes; unacceptable in Phase 1. [S30]
Auto-increase escalating into emergency tiers	Conflates growth with crisis response; could run emergency ops without a DR declaration. [D10]
Fixed-timer cooldown	Less adaptive than recovering the budget incrementally using observed success rates. [PRR §-]
No lead-time alerting (react on exhaustion)	Quota approvals take too long; reacting at exhaustion guarantees deployment failures. [FR-7.4]
Sizing without a DR buffer	Under-reserves relative to DR obligations; the formula includes an explicit <code>DR_Buffer</code> term. [FR-7.3]

Appendix A — Decision Log Cross-Reference

ADR	Primary Decisions	Hard Constraints	Key Gaps/Blockers
ADR-001 Region Selection	D1, D4, D5, D8, D9	HC-1, HC-4, HC-5, HC-8, HC-9, HC-10	G-7 (worked examples)

ADR	Primary Decisions	Hard Constraints	Key Gaps/Blockers
ADR-002 Quota & Capacity	D6, D7, D9	HC-2, HC-3, HC-7	B-1 (Quota Groups GA, POC-30)
ADR-003 Capacity during DR	D8, D10, D11	HC-1, HC-4, HC-6	G-14 (credential), G-15 (engine mode), B-2 (POC-31)
ADR-004 Forecast & Increase	D10	— (uses HC-3 at execution)	G-16 (workload tags), G-24 (increase entity)

Appendix B — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR (referenced explicitly)

Appendix C — Evidence Tag Taxonomy

Tag	Meaning
[Documented]	Traceable to Azure platform documentation or a formal FR/NFR
[Decided]	Explicit design choice in the Decision Log (D1–D11)
[Derived]	Logical consequence of a documented constraint or decision
[Assumed]	Architectural judgement pending POC validation

Document Status: Accepted

Next Review: After POC-30 (Quota Groups GA) and POC-31 (quota release latency), and on resolution of G-14 / G-15.